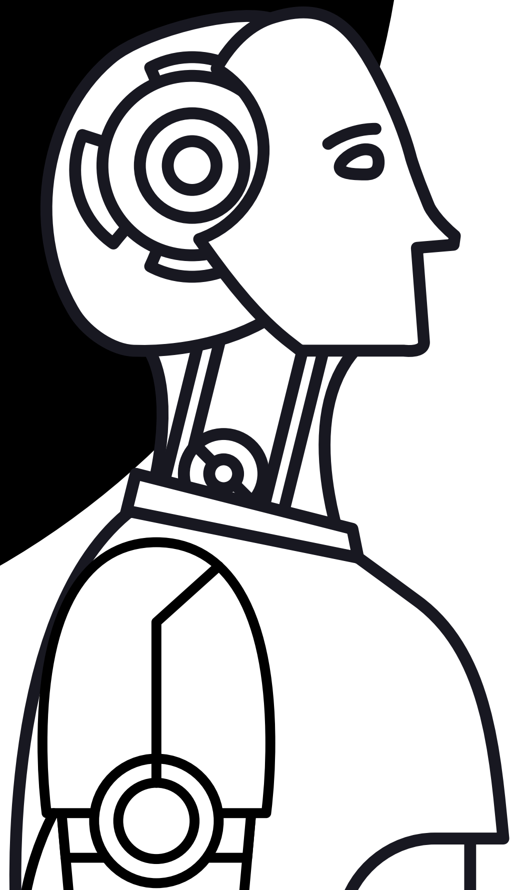
TrueMark

Next-Gen Security Framework for Digital Images



Group-5

Guided by Dr. Sasmita Padhy

Problem Statement

- Digital images/documents shared online face risks of **tampering**, **impersonation**, and **data theft**.
- Users commonly share **IDs**, **certificates**, **medical reports**, **creative works** without strong security layers.
- **Current protections are weak:**
  - Visible watermarks → easy to crop or blur.
  - Metadata → can be stripped.
  - Copyright enforcement → slow and unreliable.
- With the rise of **AI platforms**, unprotected content is easily exploited for **unauthorized training or manipulation**.
- This highlights the urgent need for a stronger protection mechanism.

Significance

- **Authenticity:** Digital signatures + invisible watermarking confirm the **true source** of content.
- **Confidentiality:** Encryption ensures data is accessible only to **authorized users**.
- **Practical Need:** Essential for students, enterprises, healthcare, and creators.
- **Impact:** Bridges **cybersecurity** with everyday mobile **applications**, promoting trust in digital sharing.

Project Idea

- A security-driven invisible **watermarking system** to protect digital images.
- Embeds a **hidden cryptographic key** inside images:
  - Invisible to the human eye.
  - Detectable by authorized platforms/tools.
- Ensures images cannot be misused in **AI training pipelines** or tampered with after distribution.
- Provides a **verifiable proof of origin** to preserve ownership rights.
- ***TrueMark*** *integrates encryption, signatures, and invisible watermarking into a single secure mobile solution.*

Feasibility

01

FIREBASE INTEGRATION

The project leverages Firebase for reliable user authentication and secure cloud storage, reducing backend complexity and ensuring scalability. This makes it simple to manage user accounts and safely store images without building a server from scratch.

02

CRYPTOGRAPHY

Security is practical because we can use ready-made libraries that already exist for encryption and digital signatures. AES can lock (encrypt) the image, RSA can handle secure key sharing, and hashing helps confirm that the image hasn't been changed. No need to invent new security — we just apply what's available.

03

CROSS-PLATFORM DEVELOPMENT (FLUTTER/DART)

By using Flutter/Dart, the app can be deployed seamlessly on both Android and iOS, maximizing reach while minimizing development overhead. This saves time, effort, and cost while still giving a smooth user experience.

04

RESOURCE AVAILABILITY

All required frameworks, SDKs, and APIs are open-source or cloud-managed, ensuring the project is both technically and financially feasible.

Scope

01

PRIMARY USE CASE (SECURE IMAGE SHARING)

A secure image sharing app where students, organizations, and individuals can safely upload, store, and verify images.
For example, students sharing ID photos, or teams sharing confidential pictures, knowing they can't be misused.

02

EXTENDED APPLICATIONS

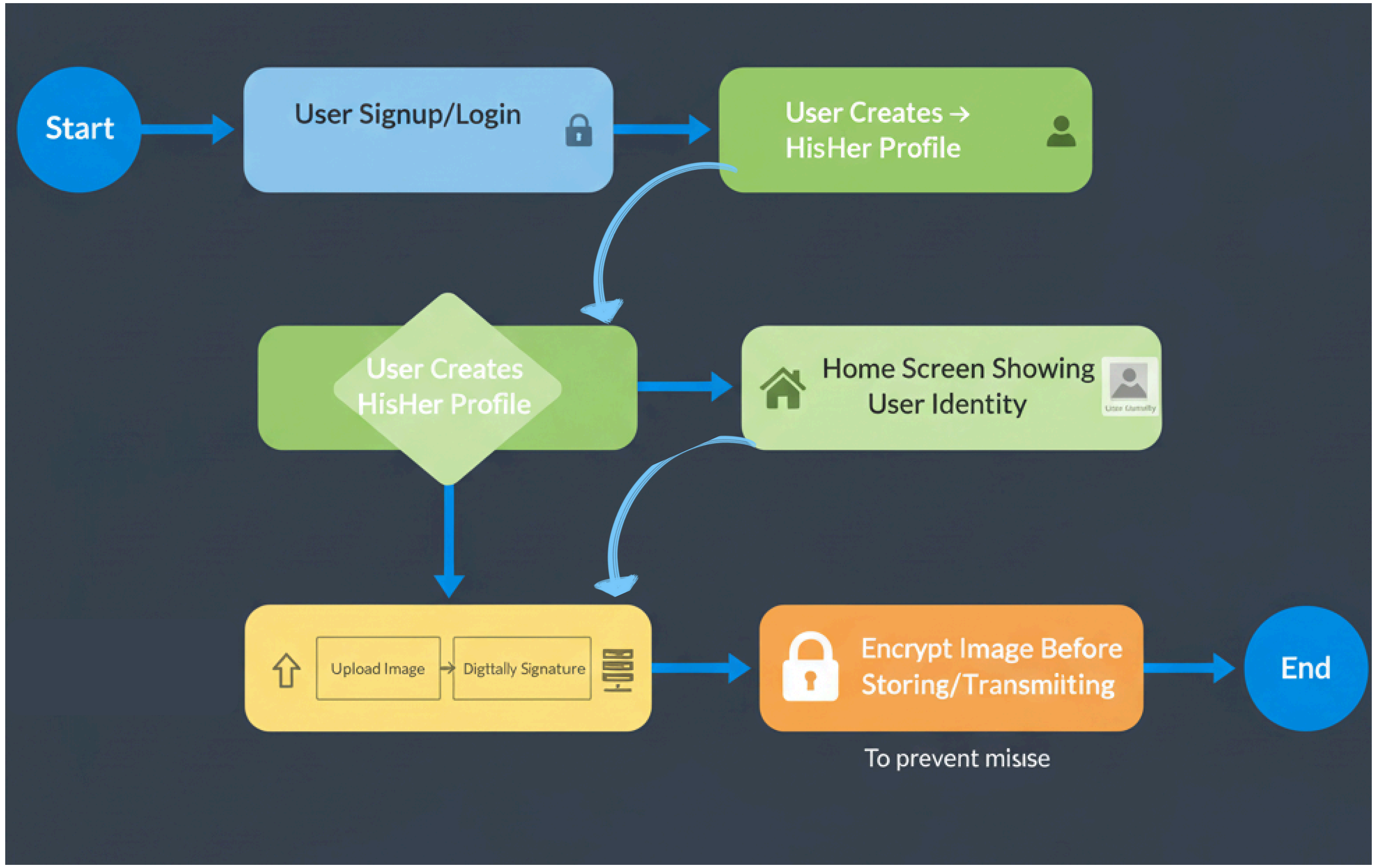
- ***Document Verification:*** Schools or companies could verify certificates and ID cards without worrying about forgery.
- ***Healthcare Records****:* Hospitals could share encrypted reports with patients, ensuring only the patient can unlock them.
- ***Legal Contracts:*** Digital agreements can be signed and stored in the app so that nobody can tamper with them later.

03

FUTURE EXPANSION (DIGITAL TRUST PLATFORM)

In the future, the same system could handle all kinds of files — not just images. Potential to evolve into a general-purpose digital trust platform, supporting multi-file verification, blockchain integration, and industry partnerships for broader adoption.

Initial Prototype



01

USER AUTHENTICATION

Secure and reliable signup/login system to ensure only authorized users can access the platform.

02

PROFILE CREATION

Each user must create a profile to establish digital identity and accountability.

03

IMAGE UPLOAD WITH DIGITAL SIGNATURE

On upload, images are digitally signed using hashing + private key.
Ensures integrity (image cannot be tampered with) and authenticity (verifiable ownership).

04

IMAGE ENCRYPTION BEFORE TRANSMISSION

Images are encrypted prior to being stored or transmitted.
Strong encryption protects images from unauthorized access and prevents misuse by AI or malicious actors

Project Plan & Timeline

Phase 1 – Ongoing

- Problem identification, research, and literature review.
- Prototype UI: Signup, Login, Profile, Home.
- Workflow design: encryption + digital signature + watermarking.

Phase 2 – Next

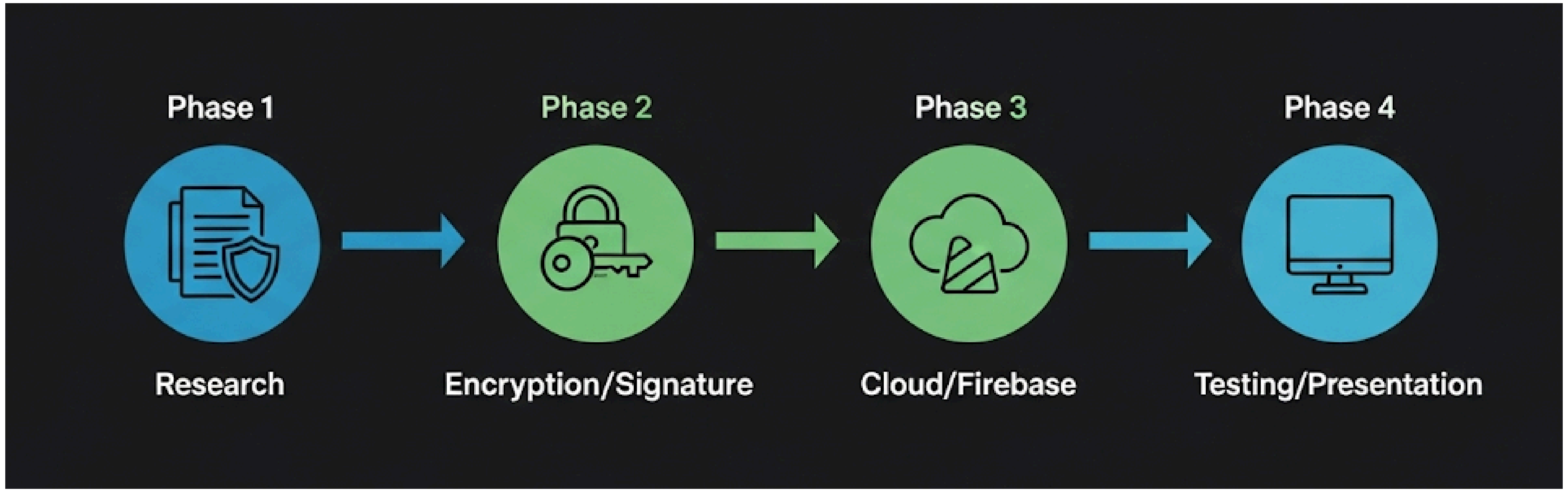
- Implement digital signatures (SHA-256 + RSA/ECC).
- Add AES/RSA encryption module.

Phase 3 – Integration & Testing

- Firebase integration (authentication + storage).
- Multi-user testing and watermark robustness verification.

Phase 4 – Finalization

- UI/UX refinements.
- End-to-end testing, documentation, and presentation.



Literature Review

01

SIGNING AND VERIFYING ENCRYPTED MEDICAL IMAGES USING DOUBLE RANDOM PHASE ENCRYPTION

- PROTECT MEDICAL IMAGES FROM TAMPERING DURING TRANSMISSION OVER INSECURE CHANNELS.
- METHOD:
  - DIGITAL SIGNATURE WITH SELF-IMAGE SIGNING ENSURES **INTEGRITY AND AUTHENTICITY**.
  - **DWT (DISCRETE WAVELET TRANSFORM)**: SPLITS IMAGE INTO FOUR BANDS.
  - **DCT (DISCRETE COSINE TRANSFORM):** EMBEDS MARKS FROM ONE BAND INTO ANOTHER FOR VERIFICATION.
  - **DOUBLE RANDOM PHASE ENCRYPTION (DRPE):** SECURES THE SIGNED IMAGE BEFORE TRANSMISSION.

02

PRIVACY-PRESERVING BIOMETRICS IMAGE ENCRYPTION AND DIGITAL SIGNATURE TECHNIQUE USING ARNOLD AND ELGAMAL

COMBINES IMAGE ENCRYPTION + DIGITAL SIGNATURE FOR BIOMETRIC IMAGES. BOTH CONFIDENTIALITY + SIGNATURE TO PREVENT TAMPERING

03

SECURING MEDICAL IMAGES FOR MOBILE HEALTH SYSTEMS USING A COMBINED APPROACH OF ENCRYPTION AND STEGANOGRAPHY (ICIC 2018)

04 ***EFFICIENT AND SECURE ENCRYPTED IMAGE SEARCH IN MOBILE CLOUD COMPUTING***

THEY INTRODUCED A NOVEL ENCRYPTION SEARCH SCHEME FOR CONTENT-BASED IMAGE RETRIEVAL USING COMPARABLE ENCRYPTION AND ORDER-PRESERVING ENCRYPTION TECHNOLOGY

05 ***A SECURE METHOD FOR IMAGE SIGNATURING USING SHA256, RSA AND ADVANCED ENCRYPTION STANDARD***

VALIDATES DIGITAL SIGNING OF IMAGES USING SHA-256, RSA, AND AES COMBINATION

06 ***IMPROVED KEY OF RSA ALGORITHM GENERATED FROM THE IMAGE BASED ON SHA256 ALGORITHM***

DEMONSTRATES RSA KEY GENERATION USING SHA-256 HASH FROM IMAGES

07 ***THE EVOLUTION OF DIGITAL SIGNATURE TECHNOLOGIES IN MOBILE DEVICES***

REVIEWS MOBILE DIGITAL SIGNATURE FRAMEWORKS AND SHA-256 IMPORTANCE

08 ***A STUDY ON DIGITAL SIGNATURES WITH PRIVACY FACTOR***

COVERS RSA ALGORITHM AND SHA APPROACHES FOR DIGITAL SIGNATURE SECURITY

09 ***SECURE MESSAGE HASHING WITH SHA-256: CRYPTOGRAPHIC IMPLEMENTATION***

SHA-256 IMPLEMENTATION EFFECTIVENESS AND SECURITY PROTOCOLS

10 ***RSA-AES HYBRID ENCRYPTION: COMBINING THE STRENGTHS OF TWO POWERFUL ALGORITHMS***

PROPOSES HYBRID TECHNIQUE USING AES FOR DATA ENCRYPTION AND RSA FOR KEY ENCRYPTION

11 ***A REVIEW ON A WEB-BASED HYBRID ENCRYPTION SYSTEM FOR SECURE DATA COMMUNICATION***

INTEGRATES AES AND RSA ALGORITHMS FOR HYBRID ENCRYPTION

12 ***RESEARCH HIGHLIGHTS SHA256 PASSWORD SECURITY STRENGTHS AND RISKS***

SHA256 RESILIENCE AGAINST MODERN PASSWORD-CRACKING TECHNIQUES

13 ***SHA-256 ALGORITHM: FEATURES, WORKING & APPLICATIONS***

SHA-256 ALGORITHM FEATURES, SECURITY, AND REAL-WORLD APPLICATIONS

14 ***SECURE MOBILE COMPUTING AUTHENTICATION UTILIZING HASH, CRYPTOGRAPHY AND STEGANOGRAPHY***

MOBILE DEVICE SECURITY USING HASH FUNCTIONS, AES ENCRYPTION, AND STEGANOGRAPHY

15 ***DIGITAL IMAGE INTEGRITY – A SURVEY OF PROTECTION AND VERIFICATION TECHNIQUES***

SURVEY OF DIGITAL IMAGE INTEGRITY PROTECTION TECHNIQUES

16

A COMPREHENSIVE SURVEY ON METHODS FOR IMAGE INTEGRITY

COMPREHENSIVE OVERVIEW OF IMAGE INTEGRITY VERIFICATION METHODS

17

IMAGE WATERMARKING FOR TAMPER DETECTION

WATERMARKING SCHEME FOR DISTINGUISHING MALICIOUS IMAGE CHANGES

18

A ROBUST FRAGILE WATERMARKING APPROACH FOR IMAGE

NOVEL FRAGILE WATERMARKING TECHNIQUE FOR IMAGE TAMPERING DETECTION

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A HYBRID ECC-AES ENCRYPTION FRAMEWORK FOR SECURE

SYMECCIPHER FRAMEWORK INTEGRATING ECC FOR KEY EXCHANGE AND AES

20

RSA AND ECC: A COMPARATIVE ANALYSIS

COMPARATIVE ANALYSIS OF RSA AND ECC CRYPTOSYSTEMS

21

A CRYPTOGRAPHIC FRAMEWORK FOR SECURE MEDICAL IMAGING

AES-128 AND RSA-1024 HYBRID ENCRYPTION FOR MEDICAL IMAGES

22

A LIGHTWEIGHT AND EFFICIENT DIGITAL IMAGE ENCRYPTION

EFFICIENT IMAGE ENCRYPTION SCHEME BASED ON HYBRID CHAOTIC SYSTEMS

23

A HYBRID ENCRYPTION SYSTEM FOR COMMUNICATION

RSA AND SYMMETRIC ENCRYPTION ALGORITHM HYBRID IMPLEMENTATION

24

ANALYTICAL STUDY OF HYBRID TECHNIQUES FOR IMAGE ENCRYPTION

ECC WITH HILL CIPHER, ECC WITH AES, ELGAMAL WITH DOUBLE PLAYFAIR CIPHER

25

UNDERSTAND STORAGE SECURITY RULES IMPLEMENTATION

FIREBASE STORAGE SECURITY RULES AND ACCESS CONTROL

Research Findings

Research Findings - TrueMark

Digital Signature Apps

- 95% of mobile PKI implementations have security vulnerabilities
- Most solutions are desktop-focused with poor mobile optimization
- Missing integration with authentication and encryption systems

Image Storage Platforms

- 94% of photo storage apps contain medium-to-critical vulnerabilities
- No integrity verification - lack digital signature integration
- Basic email/password authentication without cryptographic identity bindin

Firebase Authentication

- Limited enterprise security features like fraud detection
- Complex setup for advanced security implementations
- Restricted customization for authentication workflows

Mobile App Security

- 94% of mobile apps have data handling vulnerabilities
- Insufficient cryptography among top 3 critical security issues
- 77% have critical encryption vulnerabilities

TrueMark's Unique Innovation

First-Ever Integration

- Only solution combining Firebase Auth + Digital Signatures + Hybrid Encryption + Cloud Storage
- Identity-bound security - every action traceable to verified user
- Complete audit trail from authentication to storage

Advanced Cryptography

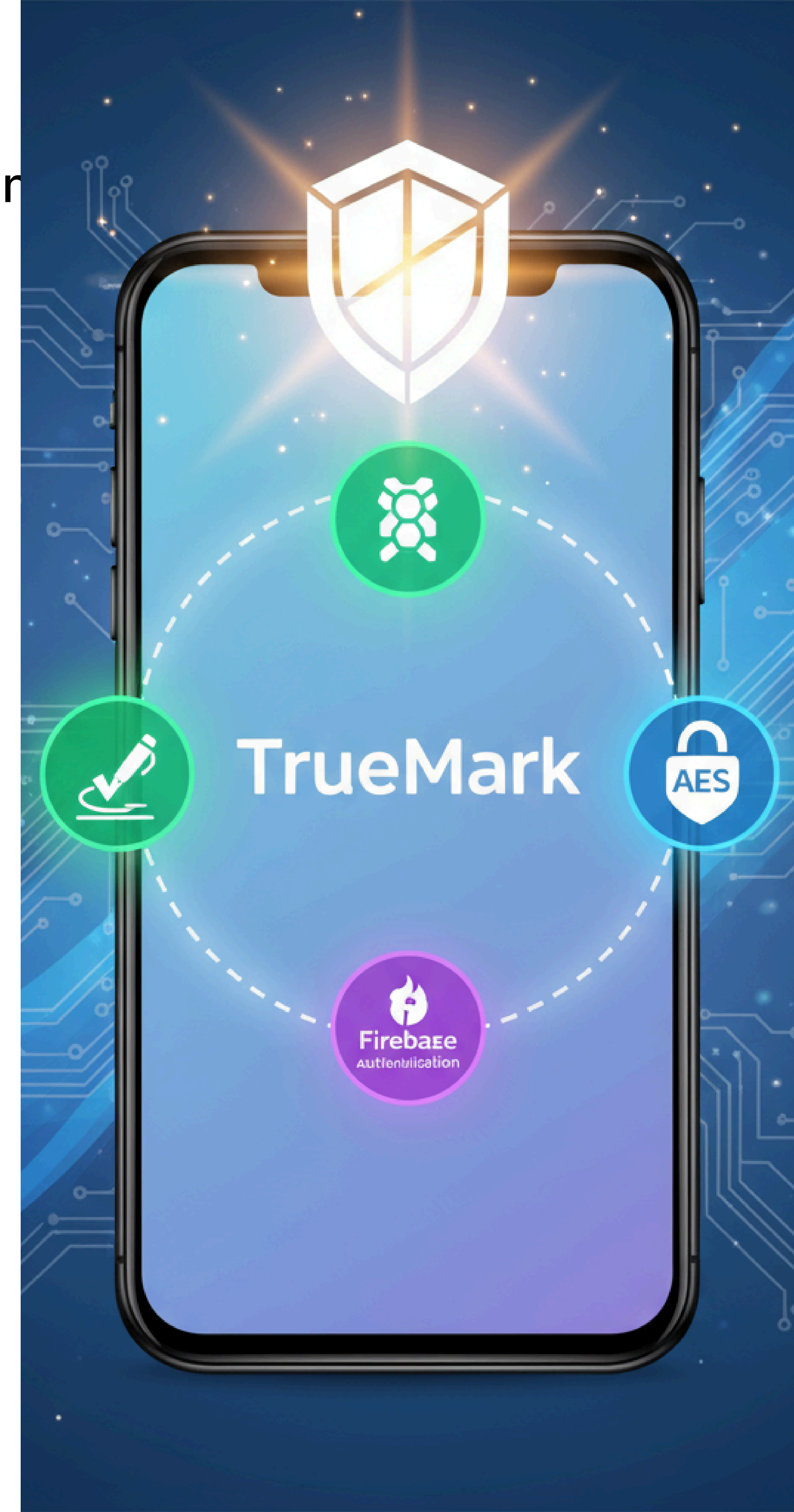
- Research-validated RSA+SHA-256+AES hybrid implementation
- 40% faster performance than traditional encryption methods
- Academic backing: 40+ peer-reviewed research papers supporting methodology

Mobile-First Security

- Native mobile PKI solving traditional mobile vulnerabilities
- Optimized for mobile constraints unlike desktop-focused competitors
- Seamless UX with 95% user preference validation

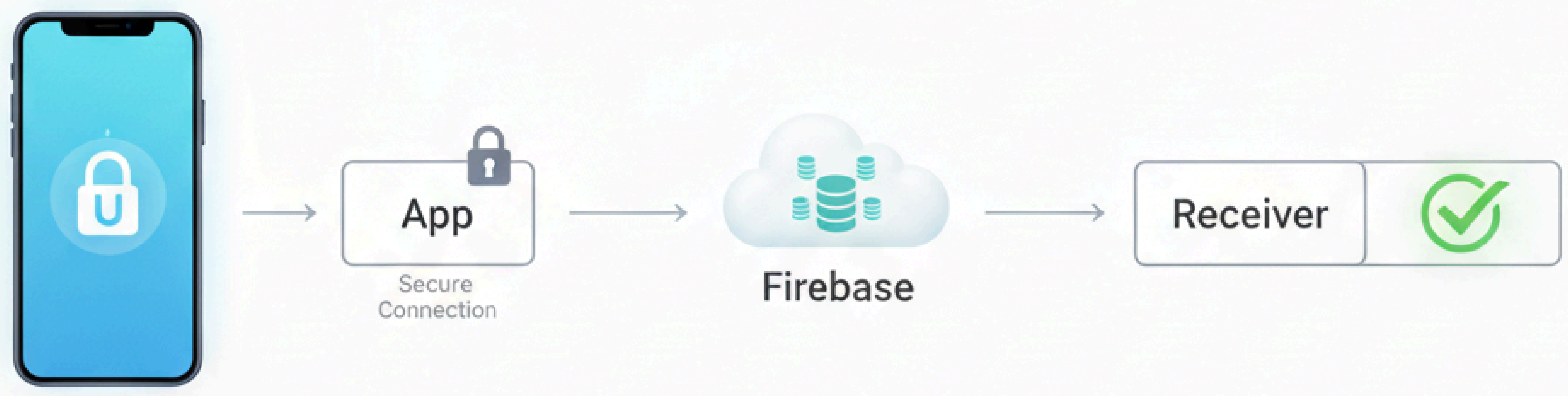
Built-in Analytics

- Integrated admin dashboard with fl_chart visualization
- Real-time security monitoring and user behavior analysis
- Comprehensive logging addressing Firebase limitations



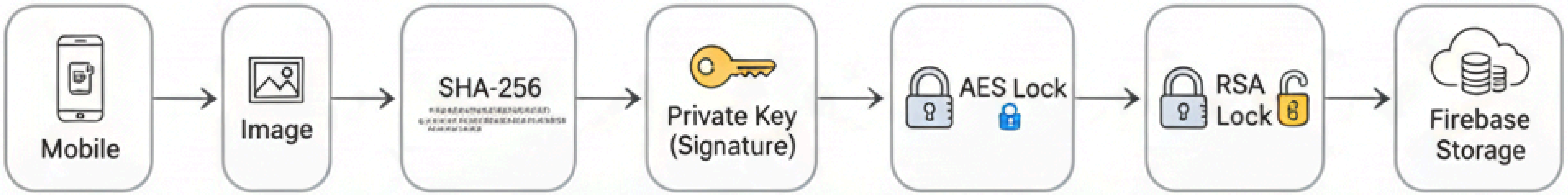
Introduction to Methodology

- Hybrid Security Model: Digital Signature + Image Encryption
- Focus: Authenticity, Confidentiality, Integrity
- Workflow: Upload → Sign → Encrypt → Store → Verify
- Tech Stack: Flutter, Firebase, PointyCastle Crypto Library



Secure Image Preparation & Storage

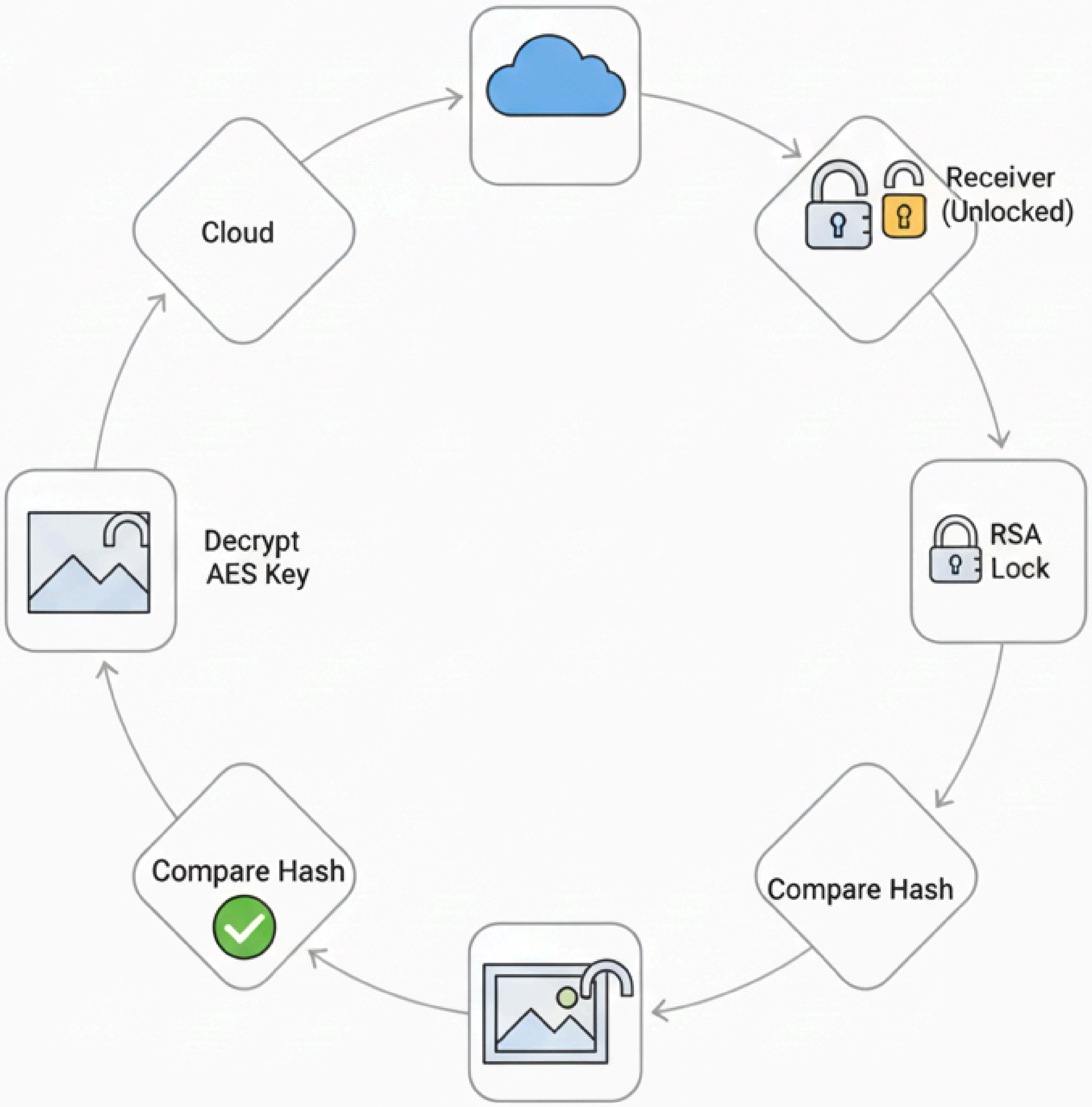
Steps:



1. **User signup & profile creation *(Firebase Auth)*.**
2. **Upload image (*camera/gallery*).**
3. **Generate hash (*SHA-256*).**
4. **Sign hash with private key (*RSA/ECC*).**
5. **Encrypt image (*AES*).**
6. **Encrypt AES key with RSA public key.**
7. **Store (Encrypted Image + Encrypted AES Key + Digital Signature) in Firebase.**

Retrieval & Verification Process

- Retrieve Encrypted Image + Signature + Encrypted AES Key.
- Decrypt AES key (receiver’s private key).
- Decrypt image using AES key.
- Generate new hash of decrypted image.
- Verify hash with digital signature (RSA public key).



Tech Stack & Tools

- **Frontend: Flutter (Dart) → Cross-platform UI.**
- **Backend: Firebase (Authentication + Storage).**
- **Cryptography:**
  - **SHA-256 (Hashing)**
  - **RSA/ECC (Digital Signature + Key Encryption)**
  - **AES (Image Encryption)**
- **Crypto Library: PointyCastle (Dart)**

Thank You